Forecasting

Instructor: Sairam Sriraman

Examiner: Prof. Dr. David Wuttke

Prof. Dr. David Wuttke

Technische Universität München

School of Management

Campus Heilbronn

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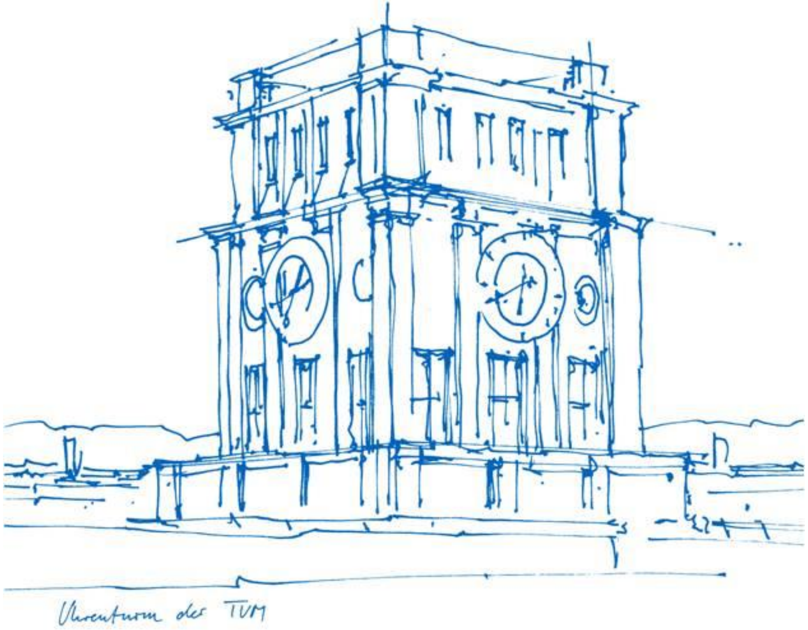
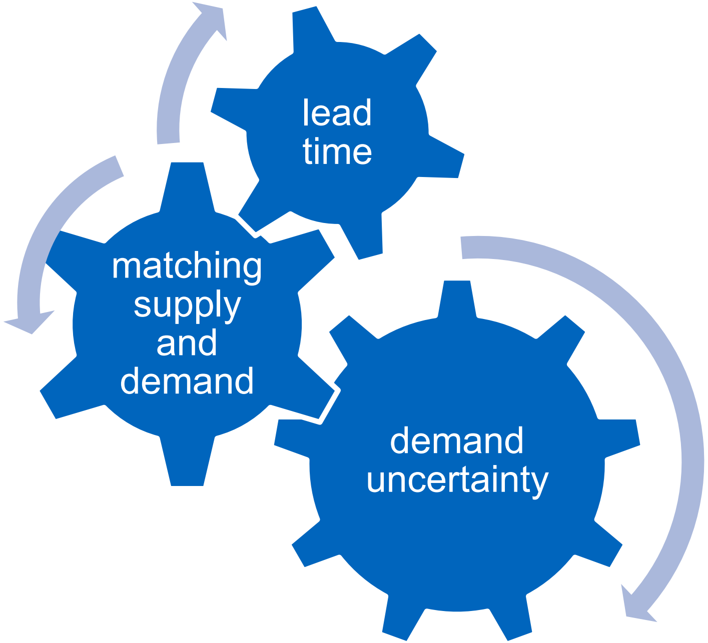


Forecasting



Organizational Process of Forecasting

Elements of a Good Forecast

- Timely
- Accurate
- Reliable
- Meaningful units
- In writing
- Simple to understand and use
- Cost-effective

Forecasting Process

Determine the purpose of the forecast

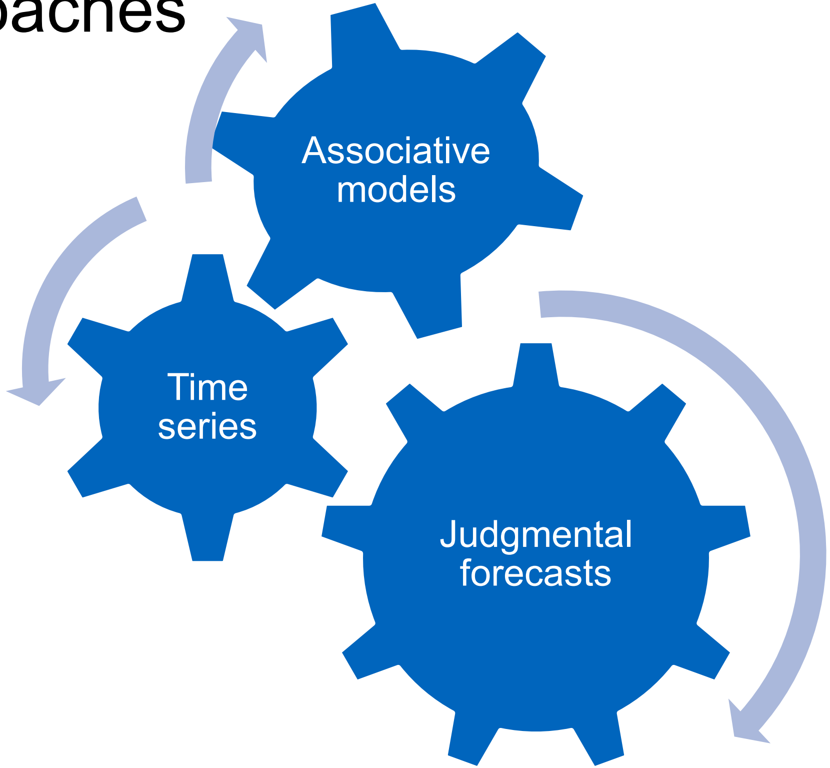
Establish a time horizon

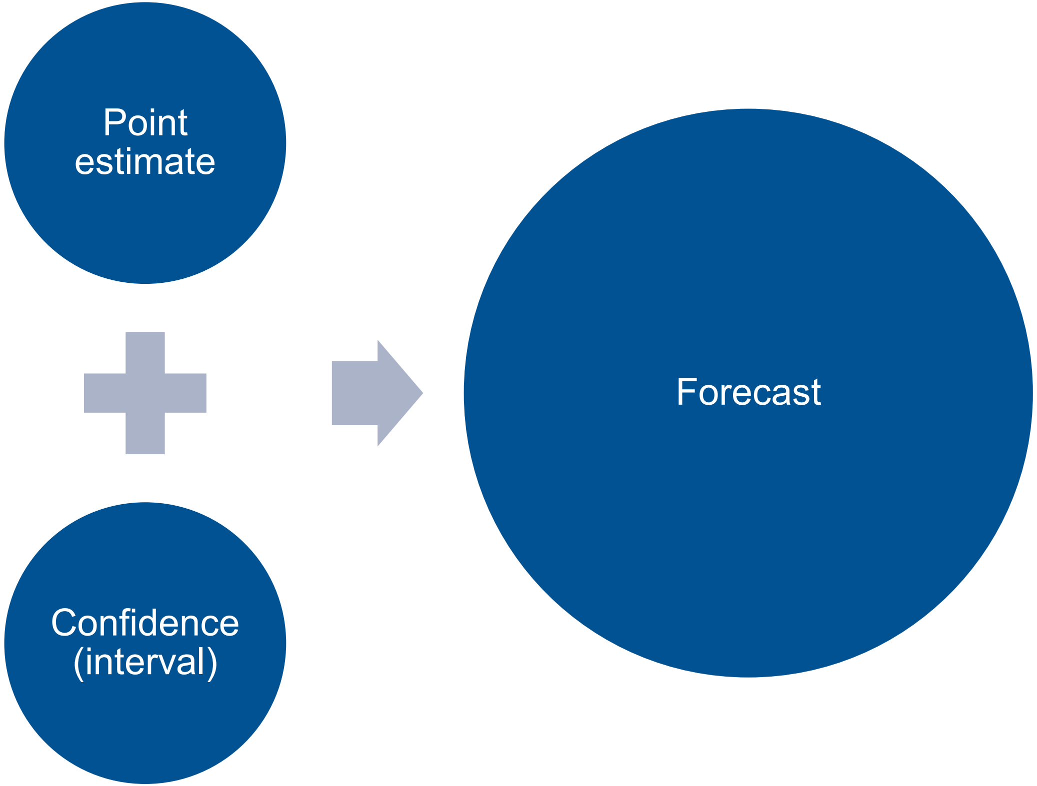
Obtain, clean, and analyze appropriate data

Select a forecasting technique

Make the forecast

Forecasting Approaches




Point estimate
Confidence (interval)
Forecast

Quantitative Evaluation Criteria for Forecasts

Time Series

> **A time series is a sequence of data points, typically consisting of successive measurements made over a time interval.**

https://en.wikipedia.org/wiki/Time_series

Time Series Example

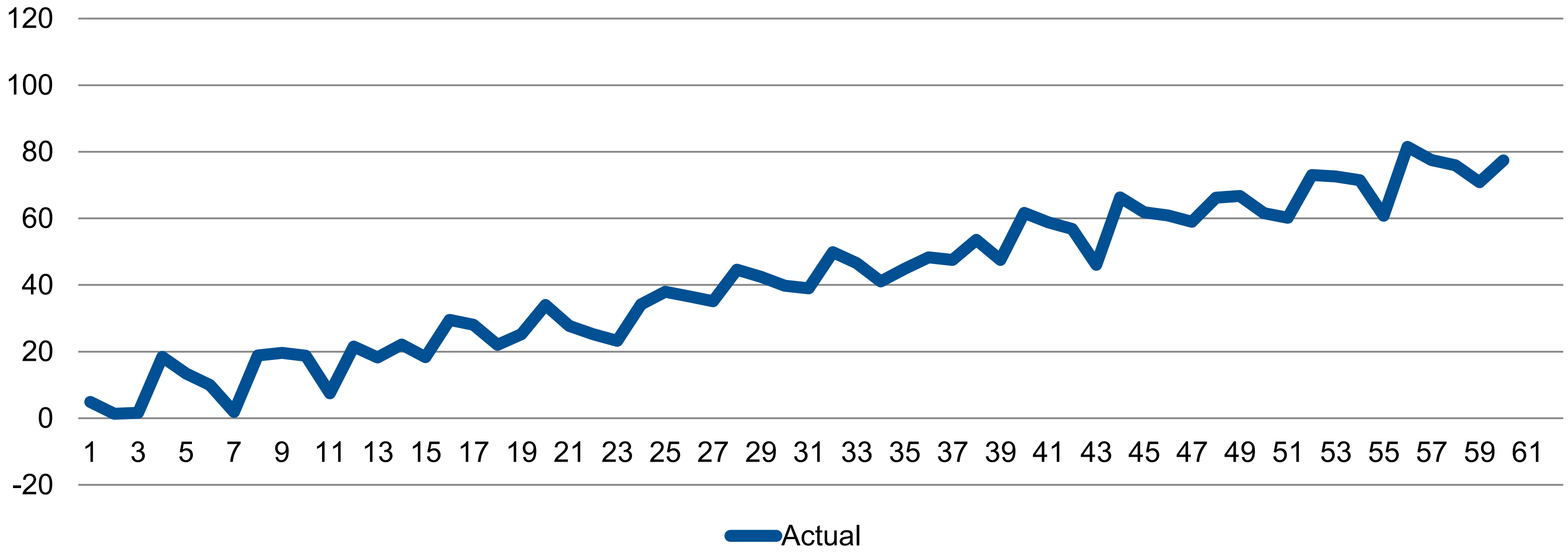
4.87, 1.34, 1.66, 18.43, 13.45, 9.99, 1.84, 18.88, 19.65, 18.75, 7.5, 21.52, 18.22, 22.11, 18.37, 29.56, 28.07, 22, 25.18, 34, 27.76, 25.23, 23.24, 34.2, 37.97, 36.63, 35.16, 44.57, 42.47, 39.79, 39, 49.89, 46.55, 41.07, 44.85, 48.34, 47.54, 53.52, 47.56, 61.64, 58.85, 56.87, 46.09, 66.33, 61.81, 60.85, 58.97, 66.22, 66.76, 61.58, 60.23, 73.04, 72.55, 71.47, 60.8, 81.48, 77.52, 75.91, 70.85, 77.41

Time Series Example

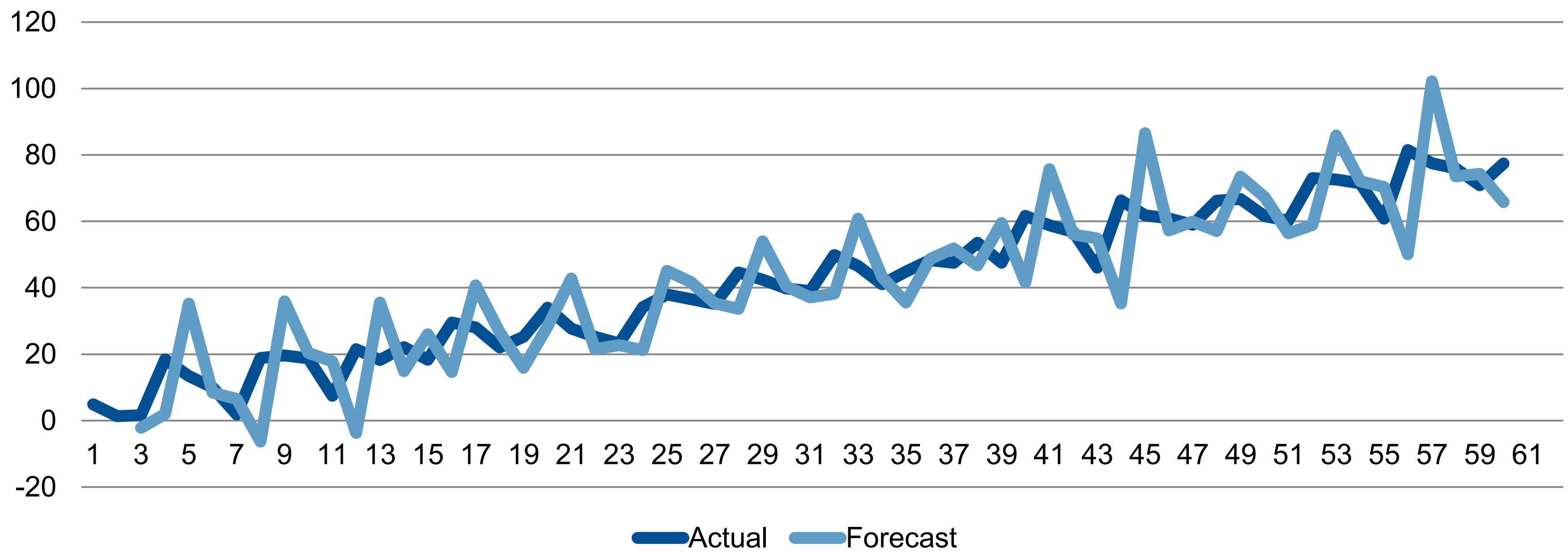
4.87, 1.34, 1.66, 18.43, 13.45, 9.99, 1.84, 18.88, 19.65, 18.75, 7.5, 21.52, 18.22, 22.11, 18.37, 29.56, 28.07, 22, 25.18, 34, 27.76, 25.23, 23.24, 34.2, 37.97, 36.63, 35.16, 44.57, 42.47, 39.79, 39, 49.89, 46.55, 41.07, 44.85, 48.34, 47.54, 53.52, 47.56, 61.64, 58.85, 56.87, 46.09, 66.33, 61.81, 60.85, 58.97, 66.22, 66.76, 61.58, 60.23, 73.04, 72.55, 71.47, 60.8, 81.48, 77.52, 75.91, 70.85, 77.41

Forecasting: Predict the next numbers (and provide confidence)!

Time Series Example: Trend, Seasonality, and White Noise



Time Series Example: Forecast



Forecasting Errors

Actual observations

- A_t

Forecast

- F_t

Error

- $e_t := A_t - F_t$

Mean Error (ME)

$$ME := \frac{1}{T}\sum_{t=1}^{T} e_t = \frac{1}{T}\sum_{t=1}^{T}(A_t - F_t)$$

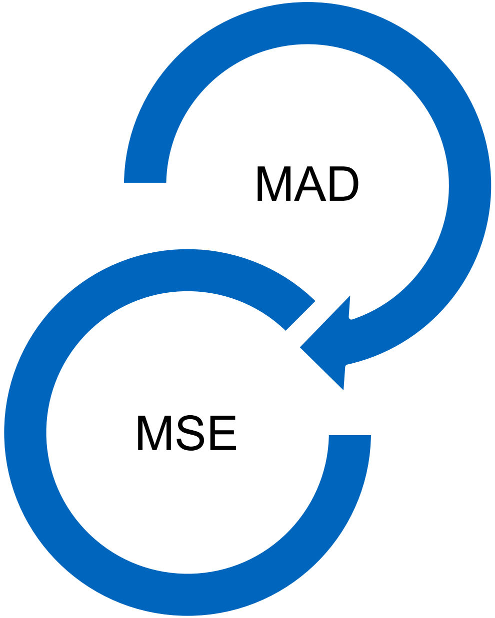
Mean Absolute Deviation (MAD)

$$MAD := \frac{1}{T}\sum_{t=1}^{T}|e_t| = \frac{1}{T}\sum_{t=1}^{T}|A_t - F_t|$$

Mean Squared Error (MSE)

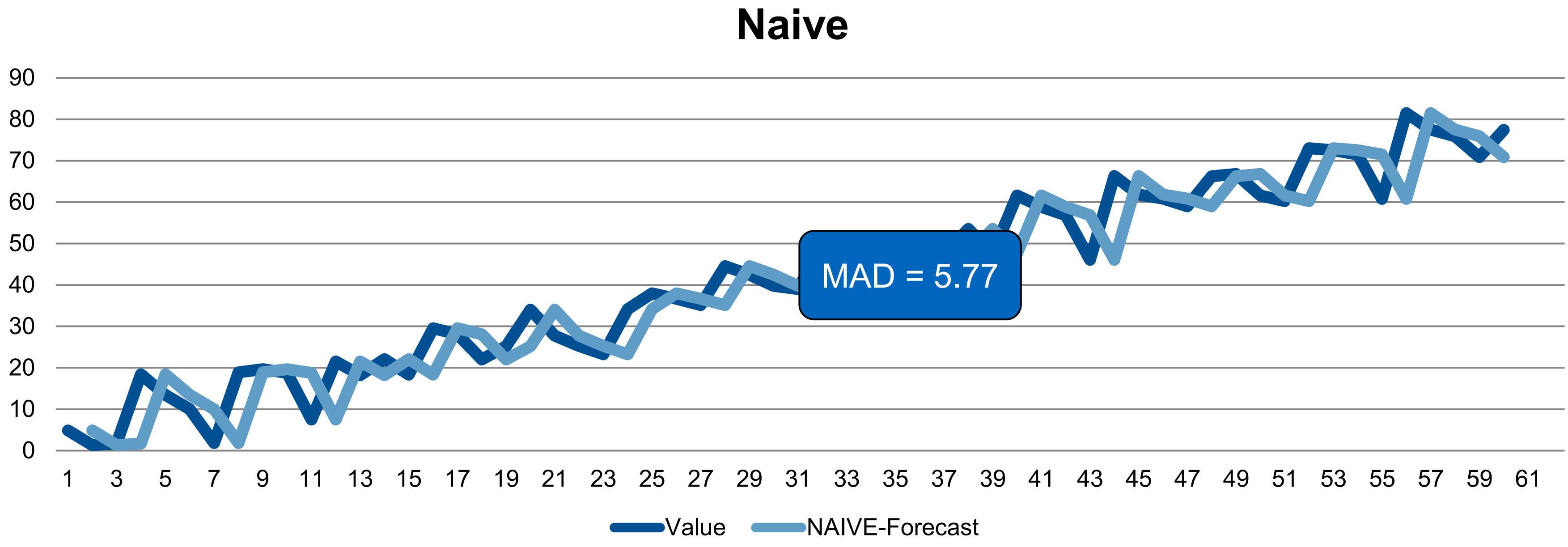
$$MSE := \frac{1}{T}\sum_{t=1}^{T} e_t^2 = \frac{1}{T}\sum_{t=1}^{T}(A_t - F_t)^2$$

Choosing Evaluation Criteria

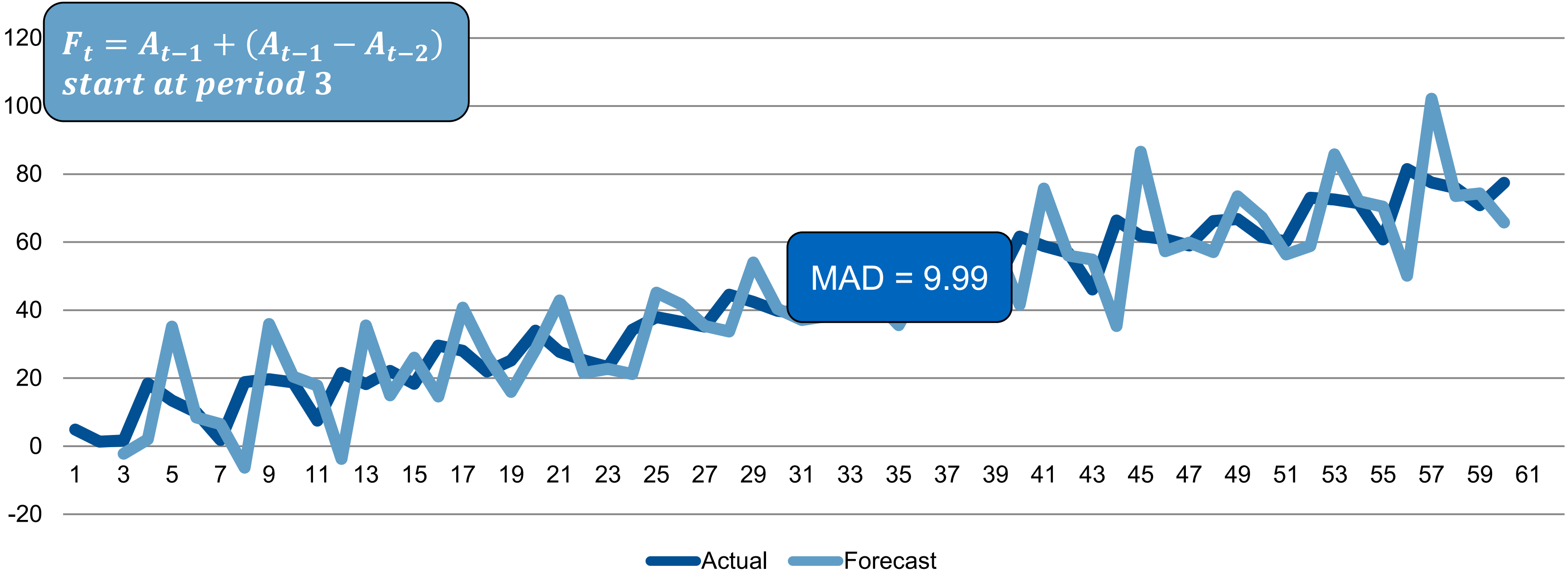


Time-Series Forecasting Methods

Naïve forecast: tomorrow the same as today

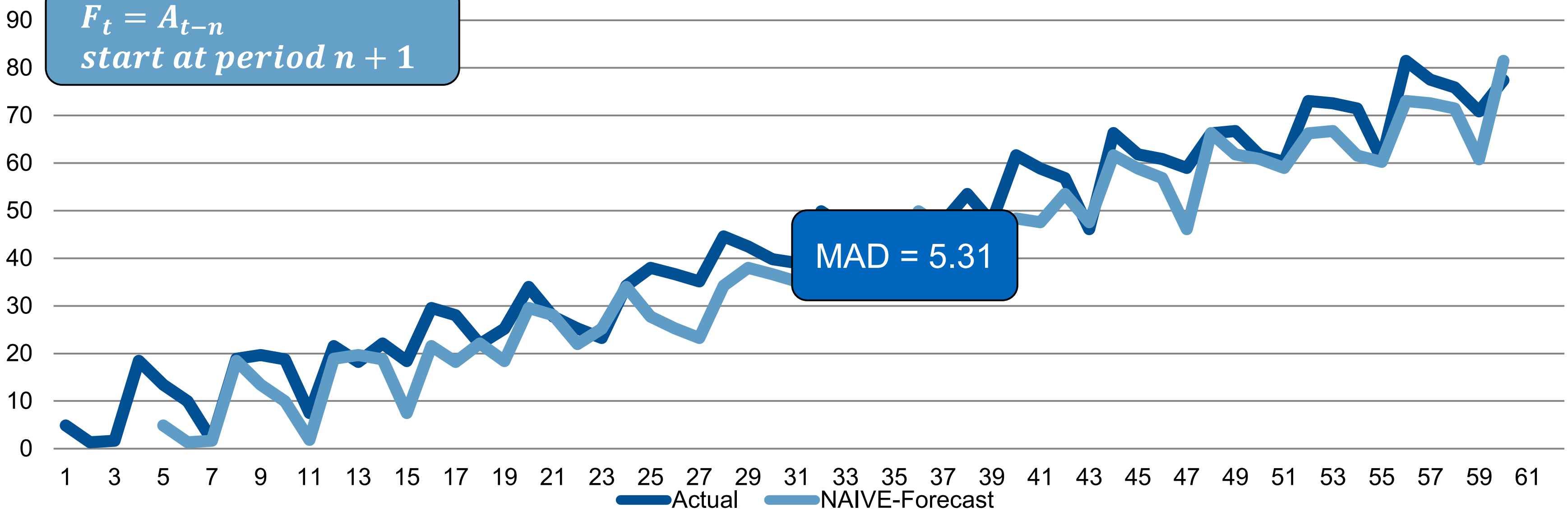


Naïve forecast with trend: same trend from yesterday



Naïve forecast with seasonality: same as last same season (in this case, 4 periods ago!)

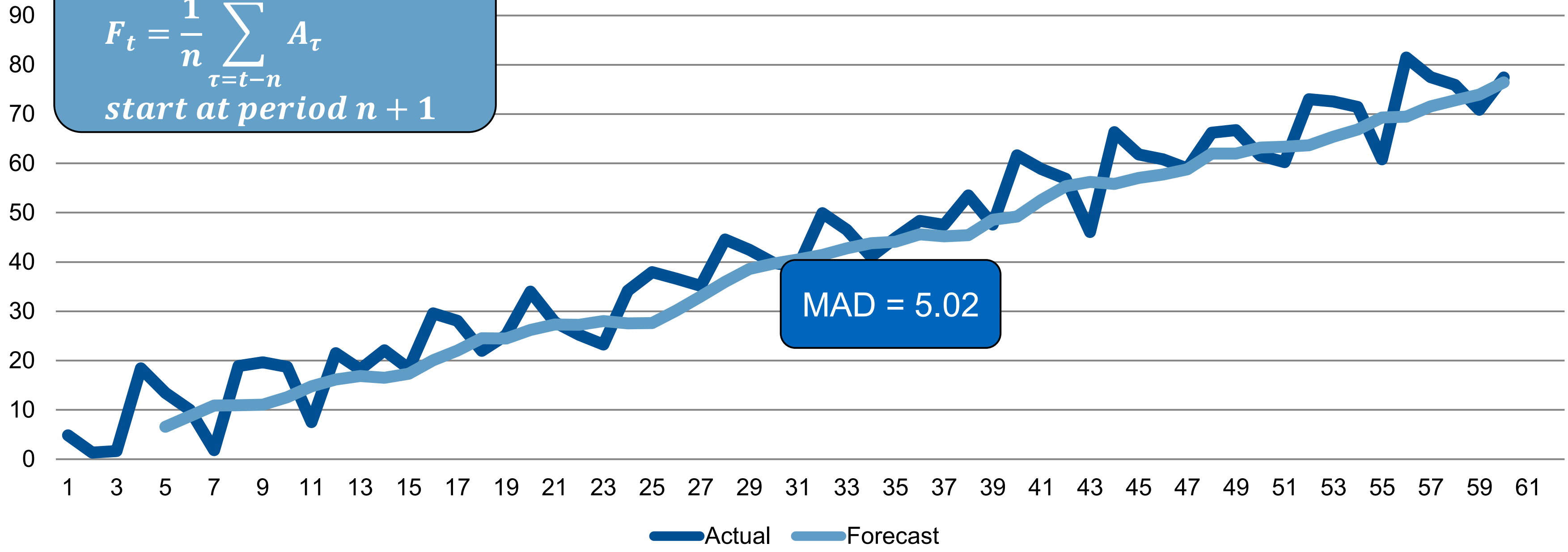
$$F_t = A_{t-n}$$
$$start\ at\ period\ n+1$$



Moving Average (same as last 4 periods)

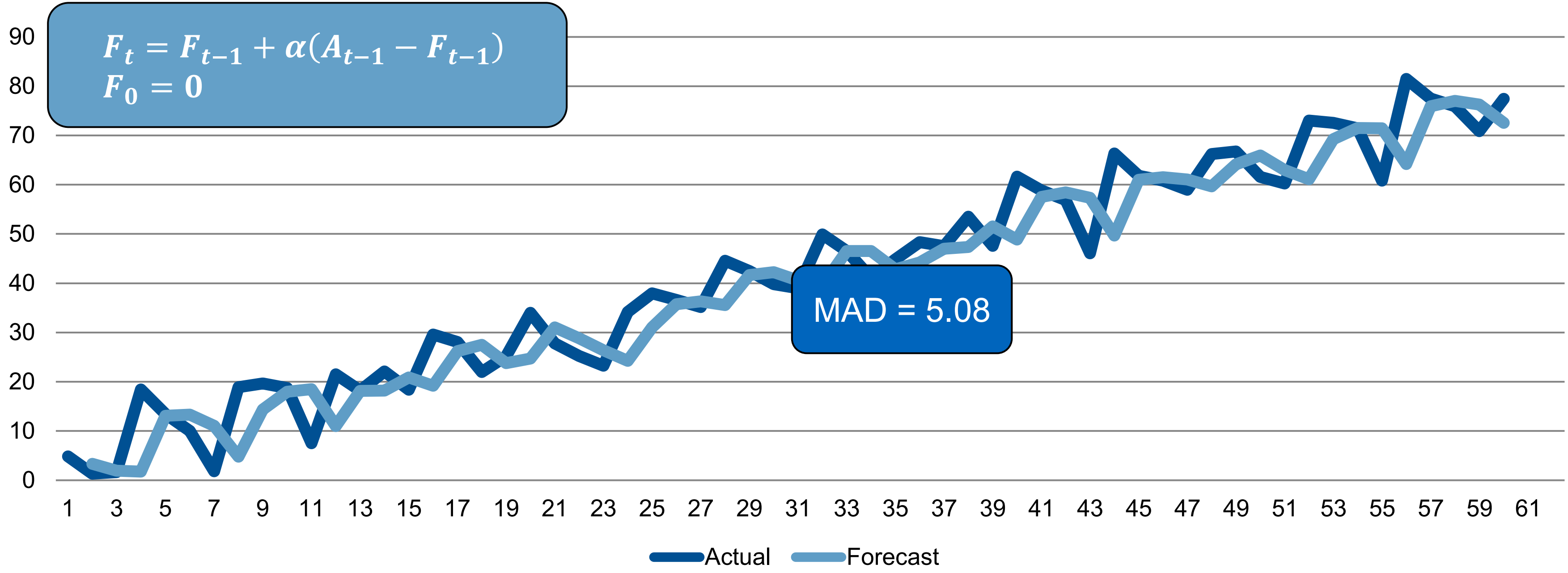
$$F_t = \frac{1}{n} \sum_{\tau=t-n}^{t-1} A_\tau$$

$$start\ at\ period\ n+1$$



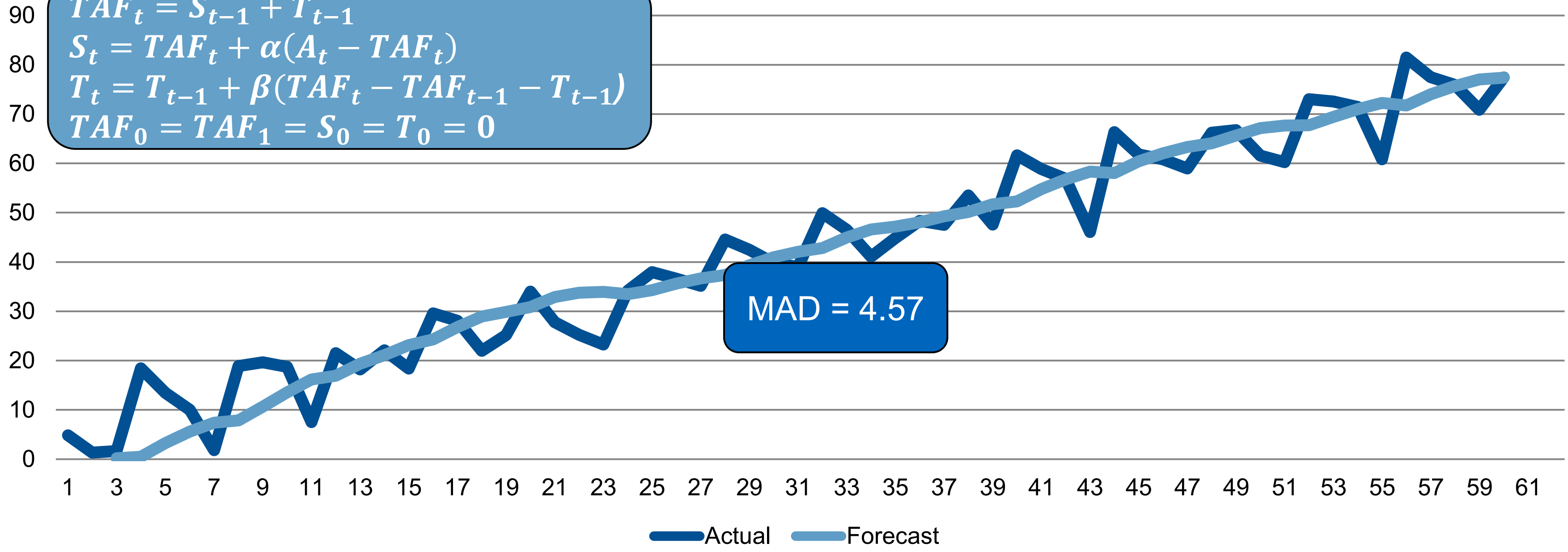
Exponential Smoothing ($\alpha = 0.68$)

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$
$$F_0 = 0$$



Trend-adjusted Exponential Smoothing ($\alpha = 0.15, \beta = 0.25$)

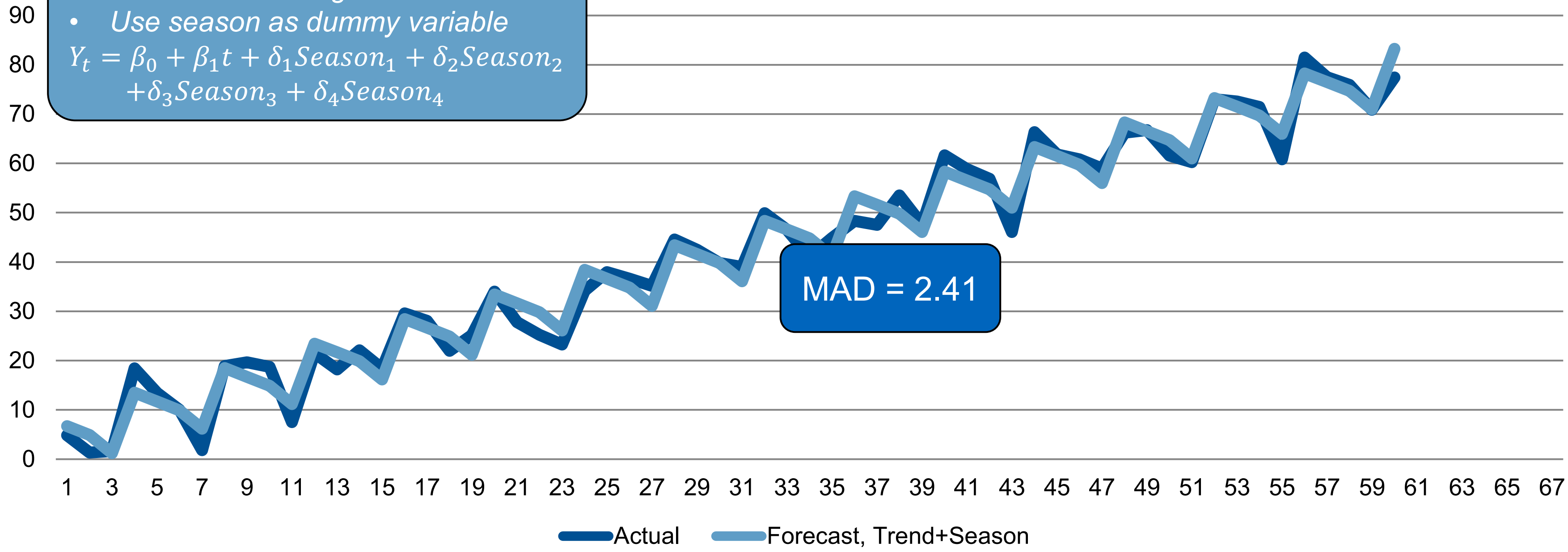
$$TAF_t = S_{t-1} + T_{t-1}$$
$$S_t = TAF_t + \alpha(A_t - TAF_t)$$
$$T_t = T_{t-1} + \beta(TAF_t - TAF_{t-1} - T_{t-1})$$
$$TAF_0 = TAF_1 = S_0 = T_0 = 0$$



Regression Model with Trend and Seasonality

- *Estimate linear regression*
- *Use season as dummy variable*

$$Y_t = \beta_0 + \beta_1 t + \delta_1 Season_1 + \delta_2 Season_2 + \delta_3 Season_3 + \delta_4 Season_4$$



Regression Model with Trend and Seasonality in R

```
A<-c(4.87, 1.34, 1.66, 18.43, 13.45, 9.99, 1.84, 18.88, 19.65, 18.75, 7.5, 21.52, 18.22, 22.11,
     18.37, 29.56, 28.07, 22, 25.18, 34, 27.76, 25.23, 23.24, 34.2, 37.97, 36.63, 35.16, 44.57,
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     46.09, 66.33, 61.81, 60.85, 58.97, 66.22, 66.76, 61.58, 60.23, 73.04, 72.55, 71.47, 60.8, 81.48,
     77.52, 75.91, 70.85, 77.41)

t<-1:length(A)

season<-as.factor(rep(1:4,length(A)/4))
df <- data.frame(t,A,season)

fit<-lm(A~t+season-1,data=df)

print(summary(fit)$coeff)

print(confint(fit, c('t', 'season1', 'season2', 'season3','season4'), level=0.95))
```

Regression Model with Trend and Seasonality in R: Output

```
            Estimate Std. Error   t value     Pr(>|t|)
t           1.243904 0.02237841 55.585011 5.129214e-50
season1     5.529450 1.00967710  5.476454 1.109398e-06
season2     2.490213 1.02420439  2.431363 1.832791e-02
season3    -2.474358 1.03901059 -2.381456 2.073234e-02
season4     8.562405 1.05408395  8.123077 5.498013e-11
```

Regression Model with Trend and Seasonality in R: Output

	Estimate	Std. Error	t value	Pr(>|t|)
t	1.243904	0.02237841	55.585011	5.129214e-50
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season4	8.562405	1.05408395	8.123077	5.498013e-11

	2.5 %	97.5 %
t	1.1990567	1.2887513
season1	3.5060120	7.5528883
season2	0.4376613	4.5427643
season3	-4.5565816	-0.3921341
season4	6.4499733	10.6748362

Comparison Across Forecast Methods

	MAD	MSE
Naive	5.77	59.10
Naive-Trend	9.99	163.46
Naive-Season	5.31	43.22
Moving Average 4 periods	5.02	36.25
Exponential	5.08	45.86
Trend-Exponential	4.57	31.91
Regression	2.41	8.23

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Monitoring Forecasts

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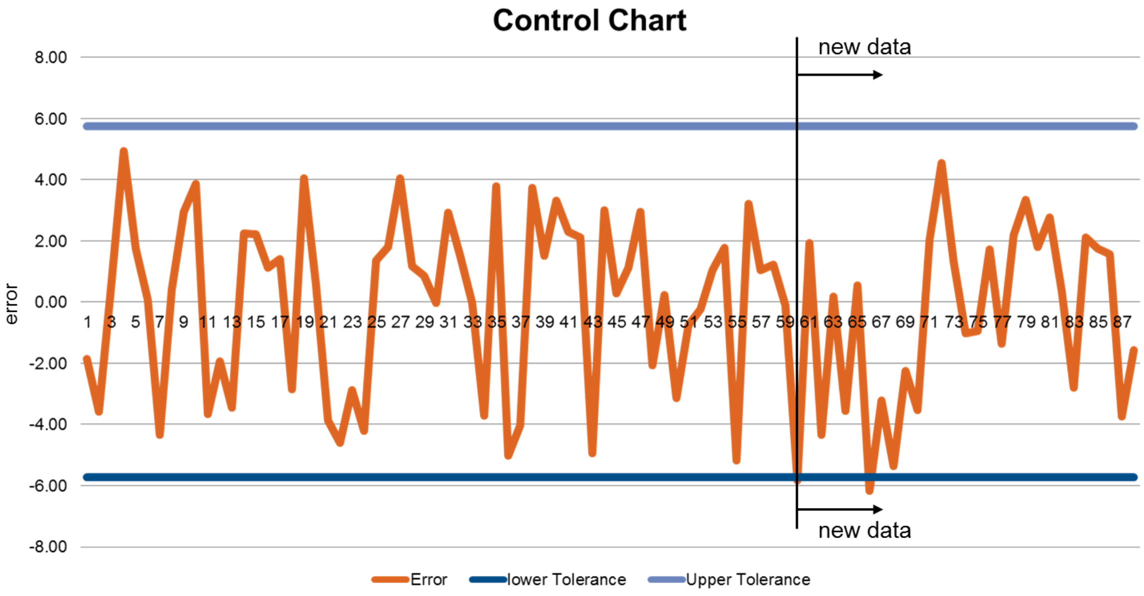
Define appropriate control limits

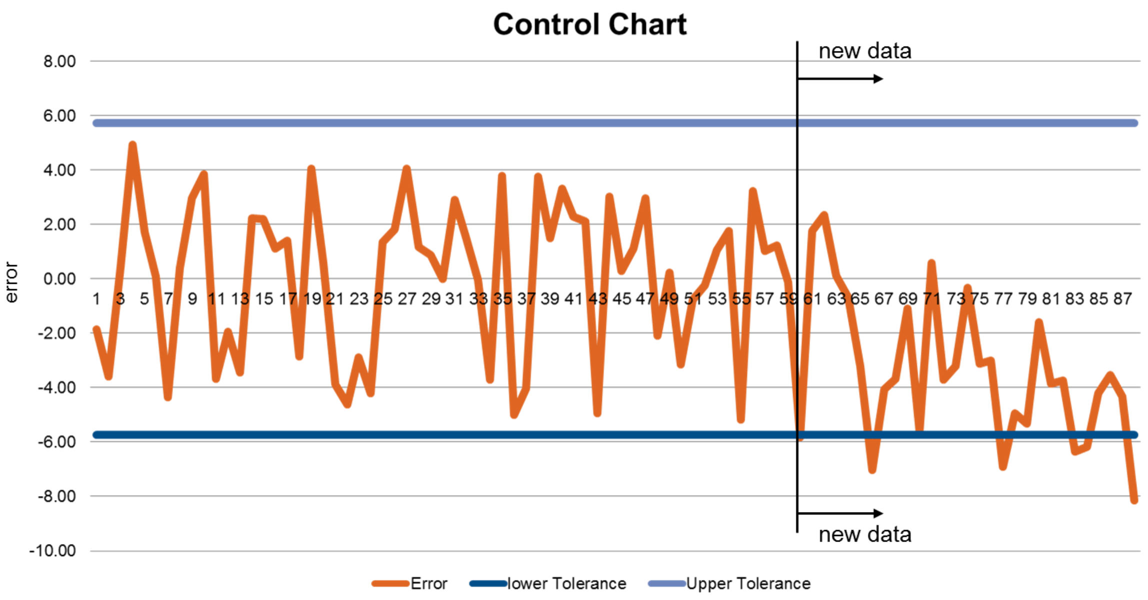
Monitor forecast errors

Respond if and only if necessary

Control Charts



Control Charts



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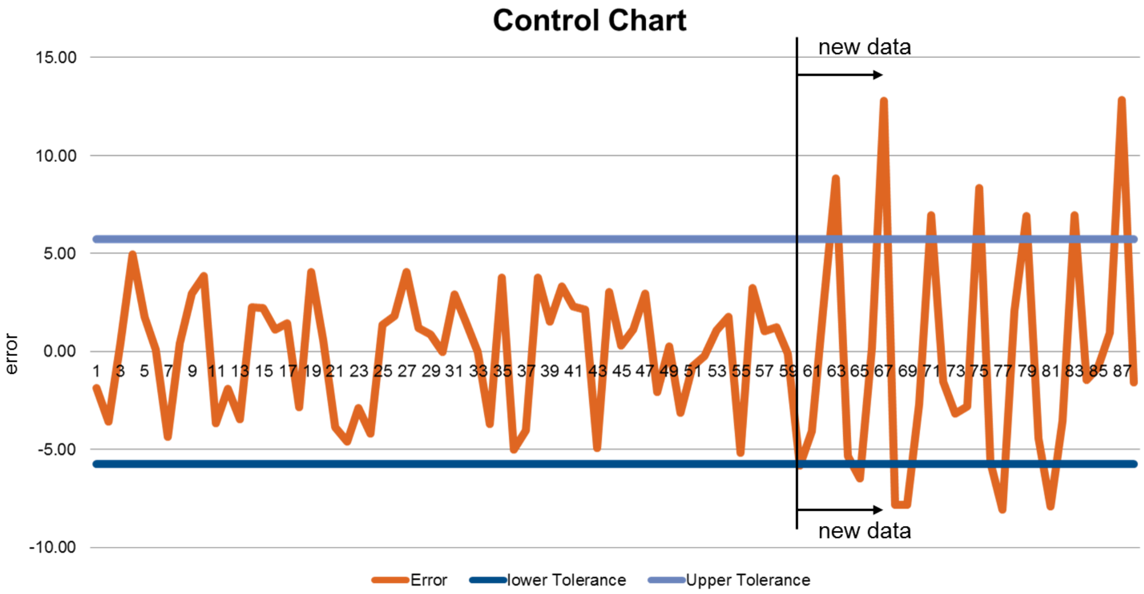

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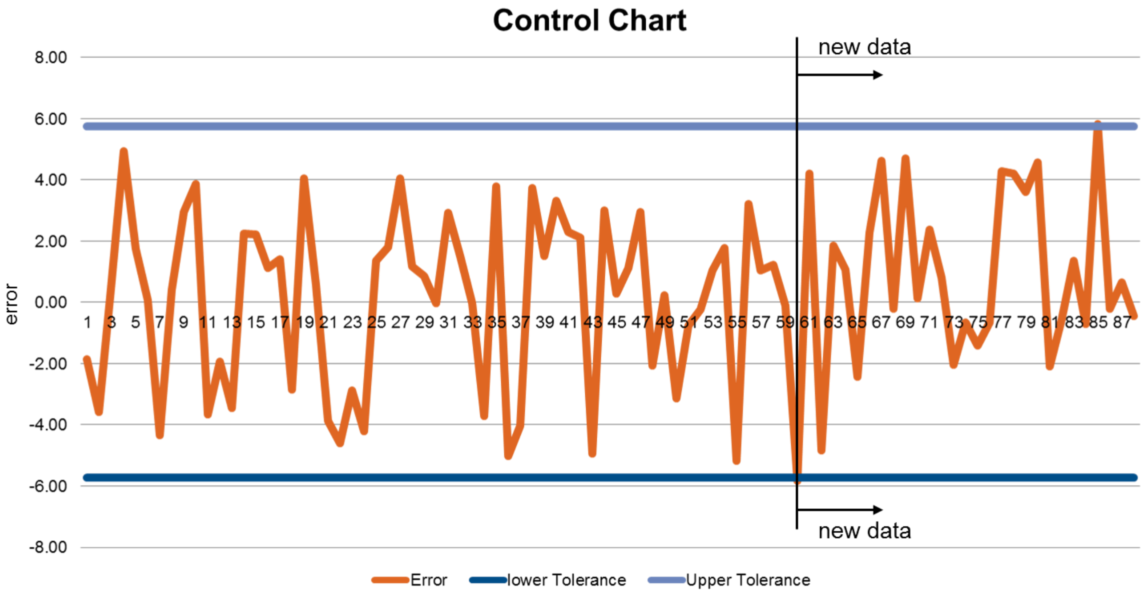

Control Charts